
STOCKTAKE: Measuring the Gap Between Perception and Action in LLM Agents with a Fair Oracle

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Abstract

LLM agents are increasingly evaluated on multi-week decision tasks in which the state that drives cost is never directly observed. On such tasks the final cost cannot say why an agent failed. It may have misread the world, or it may have read the world correctly and still failed to act (the knowing-doing gap). Existing evaluations cannot separate these two failures: their reference policies either read privileged information the agent never sees, or are missing altogether. We introduce STOCKTAKE, a 26-week supply-chain replenishment benchmark built as a factored partially observable Markov decision process (POMDP) with six hidden factor processes, designed so that a *fair* reference policy is computable: an exact Bayes filter per factor drives a rollout policy on the identical observation stream the agent receives. Scoring each run between a symptom-blind base-stock floor (0) and this oracle (1) yields a skill score, and grading each week’s written rationale yields a stated-belief detection lag and a knowing-doing rate, so state estimation and control are measured separately. On fifty seeds with curated stress profiles, Claude Sonnet 5, GPT-5.4, DeepSeek-V4-Pro, and Grok 4.5 detect 89–94% of detectable hidden failures, typically within a week of onset, yet span skill scores from 0.62 to -0.23 : two of the four fail to beat the symptom-blind floor while naming factors no slower than the two that do. The failure has two faces. Where stress persists, 18–26% of correctly diagnosed stress weeks still end in stockout (a rate that partly reflects the severity of the weeks models notice; Section 5). Under-response is only one face of the gap: a cost decomposition of every episode shows the two below-floor models bleeding instead through bulk over-ordering and holding, in excess of what the fair oracle spends on the same weeks. The bottleneck for current agents sits between a correct stated belief and the costly action it calls for; STOCKTAKE measures both directions of that failure.

1 Introduction

LLM agents are increasingly evaluated on long-horizon decision tasks: running a vending machine for months [1], managing a retail store [2], operating supply chains [3, 4]. These tasks share a structure. The world has hidden state, such as a demand shift or a failing supplier, that the agent can only infer from noisy symptoms, and performance depends on acting on those inferences week after week. Benchmarks in this genre consistently report that agent performance degrades over long horizons. What they largely cannot report is *why*.

When an agent incurs excess cost on such a task, two distinct failure modes are possible. It may have failed to *infer* the hidden state, never recognizing that demand had shifted; or it may have inferred correctly and failed to *act*, stating the correct diagnosis in its reasoning while placing the wrong order. The second failure mode is documented in bandit and game settings under the name

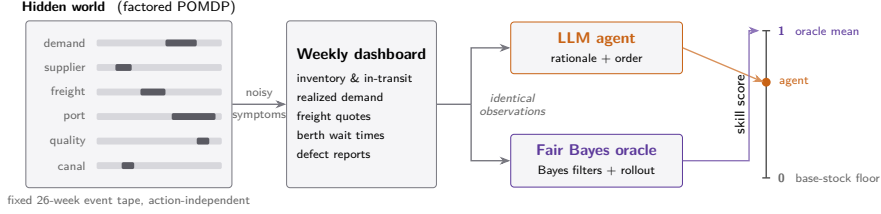


Figure 1: **Measuring the knowing-doing gap with a fair oracle.** Six hidden factor processes evolve on a fixed, action-independent event tape (left; dark segments mark stress regimes) and emit noisy symptoms into a weekly dashboard. The identical observation stream feeds both the LLM agent and a Bayes-filter reference policy (one exact filter per factor driving a rollout), so neither sees the hidden state. The agent’s episode cost is placed on a skill scale anchored by a symptom-blind base-stock floor (0) and the fair oracle’s 20-replication mean (1): shortfall on this scale is attributable to acting on beliefs, not to forming them.

of the **knowing-doing gap** [5, 6]: models often hold or state correct beliefs about the task yet fail to act on them. In the language of partially observable decision processes, these are failures of *state estimation* and failures of *control*, and they call for different fixes: better inference scaffolding in the first case, better policy elicitation in the second. Telling them apart, however, requires knowing what a correct inference *was worth*.

Existing evidence for the knowing-doing gap does not support that accounting. It comes from probes and single decisions in settings where the correct action is known in closed form [6, 5], or from cost comparisons against a reference policy that sees privileged information: the oracle in RetailBench, for instance, is computed with access to ground truth the agent never observes [2]. A privileged oracle conflates the two failure modes by construction, because the shortfall it measures bundles “did not know” and “knew but did not act” into one number. To attribute cost to the knowing-doing gap specifically, the reference policy must be denied exactly the information the agent is denied.

We introduce a benchmark in which that fair reference is computable. The task is a 26-week inventory-replenishment problem, formalized as a factored POMDP with six hidden factor processes whose states the agent must infer from noisy weekly symptoms (Section 3). Because the hidden structure is factored and the event tape fixed, an explicit Bayesian reference policy is feasible: an exact Bayes filter per factor driving a rollout policy, conditioned on exactly the observation stream the LLM receives. This reference is fair rather than privileged; Section 4 states the one disclosed asymmetry. Together with a symptom-blind base-stock floor, it yields a **skill score** locating each agent between ignoring the symptoms (0) and matching the fair Bayesian (1), and two rationale-based metrics, **detection lag** and the **knowing-doing rate**, that separate state estimation from control. Fifty seeds, grouped by stress profile (ISOLATED, PERSISTENT, COMPOUND), make the interaction of hidden failures a controlled variable.

Our contributions are:

1. We introduce a six-factor supply-chain POMDP benchmark whose upper reference is a fair, computable Bayes-filter oracle: an exact-filter rollout policy conditioned on the identical observation stream as the agent, referenced as the mean of 20 replications per seed. Fairness is what makes agent shortfall attributable to acting rather than confounded with knowing.
2. We construct a controlled stress axis (ISOLATED / PERSISTENT / COMPOUND) that varies whether and how hidden failures overlap while holding the world model fixed.
3. We measure state estimation and control separately — skill score, detection lag, knowing-doing rate — over 200 episodes: Claude Sonnet 5, GPT-5.4, DeepSeek-V4-Pro, and Grok 4.5, each on all fifty seeds, under a rules-only prompt with no strategy playbook.

Across 200 episodes, detection proves nearly uniform while costs diverge sharply; Section 5 gives the numbers. What separates models is the distance between seeing a problem and acting on it, and the failures run in both directions: ordering too little too late, and paying too much for protection.

2 Related Work

Evaluating LLM agents as sequential decision-makers has grown from single-turn tool use into month-scale simulations of running a business. Vending-Bench [1] has agents operate a vending machine across two thousand messages and finds coherence collapse, ruling out context exhaustion as the cause. RetailBench [2] simulates a supermarket built on real retail records, frames it as a partially observable decision process, and finds that only a few models survive its 180-day horizon. In inventory management specifically, AIM-Bench [3] finds human-like ordering biases (demand anchoring, bullwhip amplification) across five environments, and InvAgent [4] reports outcome gaps against heuristic and reinforcement-learning baselines in a fully observed twelve-period setting.

A second line of work asks whether agents act on what they know. In bandit tasks, models produce a correct rationale for 87% of decisions and still take the greedy action over the optimal action they just derived [5]. Probing studies in strategic games find internal beliefs about the opponent that are often accurate, weakly verbalized, and weakly linked to the chosen action [6]. Probes of tool-use decisions find the same split inside the network: the belief that an action is needed and the action taken are both linearly decodable, but their directions drift apart in later layers [7]. The phenomenon has a name, the knowing-doing gap, and it has so far been measured on decisions whose correct answer is known in closed form.

A third line evaluates the beliefs themselves. BayesBench [10] scores posterior updating against a rational Bayesian reasoner and finds systematic departures, on estimation tasks with no actions attached. The machinery for connecting such beliefs to inventory decisions is classical: Bayraktar and Ludkovski [11] derive the exact Bayes-filter reduction of inventory control with hidden Markov-modulated demand to a fully observed problem, in their case to compute optimal policies.

The task is causal only in a limited sense. Causal-reasoning benchmarks test whether a model can recover causal structure or answer interventional and counterfactual queries [8], and there is evidence that models recite causal facts from text rather than infer them [9]. Our task tests neither: the causal structure of the world is given to the agent in the prompt, and actions never touch the hidden dynamics, so there is nothing to discover and nothing to intervene on. What the weekly decision demands is abduction, inferring which hidden cause best explains the week’s noisy symptoms under a known model, followed by cost-sensitive control. The benchmark is complementary to causal-reasoning evaluations: it measures whether a correct causal attribution, once formed, survives the trip to an action.

Table 4 (Appendix A) summarizes the benchmarks closest to ours. In each, the reference policy either reads information the agent is denied (RetailBench’s rule-based oracle takes “structured simulator fields” as input; AIM-Bench’s ex-post optimum is computed from realized demand) or is absent beyond a single human run, and none grades the agent’s stated beliefs alongside its actions. STOCKTAKE addresses both limitations with one construction: the classical filter reduction, repurposed as a reference policy conditioned on exactly the agent’s observation stream, so that shortfall against it isolates the action side of the knowing-doing gap, while the weekly stated beliefs it is compared against are graded on the same episodes.

3 STOCKTAKE: Task and Environment

STOCKTAKE casts the agent as the replenishment manager of an electronics importer for a 26-week episode. The state that drives cost is never observed directly. The agent reads weekly symptoms, commits an order, and writes down what it believes is happening (Figure 1).

3.1 The weekly decision

Each week the agent reads a dashboard of on-hand inventory, orders in transit, realized demand, freight quotes, berth waiting times, canal transit counts, supplier scorecards, and quality reports. It may then take any of five within-week actions, none of which advances time. `lock_freight` fixes the current freight multiplier for a chosen number of weeks; the lock itself is free, but a locked rate forgoes any drop in the spot index. `expedite_air` flies up to 20 units in at \$15 per unit; they land the following week and bypass the port entirely. `inspect_batch` (\$40) sorts an arriving batch so

that most defective units are recovered before they stock. `buy_briefing` (\$30) and `buy_audit` (\$25) reveal the true state of the canal and of the spot supplier, respectively.

The week is then committed by exactly one call to `place_order(rationale, qty, route, supplier, contract)`. The order names a quantity, a route (through the Suez Canal, or slower and dearer around the Cape), and a supplier; the optional contract action signs, switches, renews, or lapses a supplier contract on short, long, strict, or lenient terms, and the agent can only source suppliers it holds a live contract with. The `rationale` argument is required, and the world does not advance without it. Every run therefore contains 26 written statements of what the agent believed each week; Section 4 derives the belief-side metrics from them.

Costs accrue for units held and in transit, for demand missed, and for the priced levers above; the episode score is total cost, lower is better. The price ladder is the task’s core tension: sea freight is \$4 per unit and takes three weeks, air is \$15, and every unit of missed demand costs \$20. An agent that reads the symptoms early can buy its way out of trouble cheaply; one that reads them late cannot.

3.2 The hidden world

The environment is a factored POMDP. Two structural properties of its hidden state make a fair reference policy computable. Let $x_t = (x_t^1, \dots, x_t^6)$ denote the hidden state in week t , one factor process each for demand regime, supplier health, freight market, port congestion, batch quality, and canal status, each a small Markov chain over regime–age pairs that moves between a baseline and one or more stress regimes (diagrammed in Figure 3, Appendix B). The regime and its age set the transition probabilities: how likely a stress episode is to start, to persist once started, and to resolve.

First, the dynamics are factored and action-independent. Each factor steps as $x_{t+1}^i \sim P^i(\cdot \mid x_t^i)$, reading neither the other factors nor the agent’s action; the entire generative process is one random stream, seeded once, consumed by the six factors in a fixed order (Algorithm 1, Appendix B). The seed therefore fixes every regime change and every noisy reading for all 26 weeks before the agent exists; we call this fixed record the episode’s *event tape*. Two policies run on the same seed face byte-identical shocks and symptom noise, so the reference policies of Section 4 are evaluated on the very tape the agent lived through. Actions decide how much each shock costs, never what happens in the hidden world.

The port factor instantiates the template. From clear, the port can start building toward a backlog or take a sudden customs hold; building tips into congested or clears; congested persists week to week until its age cap forces recovery. Each week the realized berth wait is drawn around the current band’s mean, roughly one day when clear and sixteen when congested, and while congested every arriving shipment is held an extra week. The emission bands are where the benchmark’s ambiguities are constructed: for several factors the first week of two different regimes maps to the same band, so a fresh customs hold and newly tipped congestion both read “slow”, and a demand promotion and a seasonal lift both open on the same “surge”. A single reading cannot say which stress has begun; only the following weeks separate them.

Second, the observations are factored. The weekly dashboard o_t contains the fully observed logistics state (on-hand stock, orders in transit, realized demand) and one symptom channel per factor, $y_t^i \sim Q^i(\cdot \mid x_t^i)$. The agent never observes any x_t^i directly; the paid briefing and audit of Section 3.1 are the only exceptions, each revealing one factor’s current state for a fee. The channels themselves are hard to read in three different ways. Most are noisy: a building port and a clear one differ by a few days of berth wait, against scatter of the same size. The supplier scorecard is biased, reading one severity band better than the truth by design. The canal channel is exact but ambiguous, emitting the same transit-count crash for a false alarm as for the first week of a real closure, so even a Bayesian observer *should* be uncertain there.

The cost function, by contrast, is *not* factored. The factors evolve independently but bill jointly through a single set of books: a port hold delays the same shipments a demand surge is waiting on, and a quality escape shrinks the same inventory that surge is draining. Concurrent shocks can therefore cost far more than the same shocks arriving in sequence, and Section 3.3 samples that regime deliberately. Full cost constants, caps, lead times, and per-factor transition parameters are in Appendix B.

3.3 Seeds and stress profiles

Because the tape is a pure function of the seed (Algorithm 1), instances can be minted and classified before any agent exists. Any unsigned integer names a candidate: the world is replayed to week 26 with empty actions, and the resulting tape is summarized by a handful of features — the number of stressed weeks per factor, how many weeks have two or more factors stressed at once and the most stressed simultaneously, the longest unbroken run of port blockage, and whether the spot supplier failed outright. Thresholds on these features classify the candidate’s stress profile, and candidates too benign to separate good decision-making from bad (those on which the two reference policies of Section 4 cost nearly the same) are discarded. The benchmark fixes fifty seeds chosen this way from a 200-seed pool:

- ISOLATED (12 seeds), short single-factor shocks: no week on the tape has two factors stressed at once;
- PERSISTENT (15 seeds), a port blockage of at least four consecutive weeks alongside at least four weeks of demand stress. Here the intuitively appealing response, freezing orders until the congestion clears, is the costly one, because orders placed *through* the congestion arrive as it clears;
- COMPOUND (23 seeds), everything else: two or more concurrent factor episodes with no single blockage dominating the tape.

Compound stress is deliberately oversampled relative to the pool; it is the regime the benchmark targets, and all results are reported per group. The fifty seeds were curated iteratively across three rounds, screening for stress coverage and headroom, and the persistent-profile blockage threshold was relaxed from seven weeks to four between rounds; a committed classifier reproduces all fifty labels from the tape features alone, so new instances of any profile can be minted and classified with no agent runs and no human labeling.

3.4 Agent prompt

Agents run under a single *rules-only* prompt that specifies the mechanics of the task (costs, lead times, levers, output format) and gives a qualitative description of the world’s factors, with no strategy advice, no playbook, and no worked examples. The benchmark thereby measures what a capable agent brings to the task on its own. The full prompt and the weekly observation format are reproduced in Appendix F.

4 Experimental Setup

Reference policies. Every run is scored between two reference policies evaluated on the same seed, and both are functions of the agent’s own observation stream $o_{1:t}$, nothing more. The floor is a textbook base-stock policy: order up to a fixed target $S = \mu L + z\sigma\sqrt{L}$ (mean demand μ , lead time L , safety factor z from the holding/stockout cost ratio) every week, always by the same route and supplier, reading no symptom channel y_t^i at all. The upper reference is the Bayes-filter oracle of Figure 1. Because the dynamics and the observations both factor and neither reads the action (Section 3.2), the posterior over the hidden state splits into six independent parts, each small enough to maintain *exactly*: one forward (predict–correct) filter per factor tracks $b_t^i = \Pr(x_t^i | y_{1:t}^i)$, updated each week on the same observation dictionary o_t that is serialized into the LLM prompt. To act, the oracle samples ~ 200 possible futures of all six factors from the belief $b_t = (b_t^1, \dots, b_t^6)$, scores a small menu of candidate actions (order sizes, routes, each mitigation lever, gated on the relevant posterior risk) against every sampled future, and executes the cheapest candidate through the same action API the LLM uses. Its randomness is independent of the world’s, and it never reads hidden state. One asymmetry is disclosed: its filters use the environment’s true generative parameters (the transition tables P^i and the regime means behind Q^i), which the LLM receives only as a qualitative description. We therefore interpret the oracle as Bayes-optimal given the true model. Because it is a stochastic policy, its per-seed reference value is the mean of 20 replications (standard error 0.2–1.8% of the reference cost); it is a fair reference rather than an optimum, and agents exceed it on some tapes (Section 5). Appendix C specifies both policies in full.

Table 1: Skill score by stress profile: group mean (median), single run per model–seed pair, every model on all fifty seeds (11/15/23 scoreable per group after the seed with negative headroom is excluded). *neg* counts scoreable seeds below the base-stock floor; *beats floor* counts every seed, including the excluded one, on which the agent’s raw cost beat the floor’s.

Model	ISOLATED	PERSISTENT	COMPOUND	All	neg	beats floor
Claude Sonnet 5	0.54 (0.71)	0.70 (0.61)	0.33 (0.35)	0.49 (0.45)	9	40/50
GPT-5.4	0.21 (0.07)	0.55 (0.61)	0.86 (0.73)	0.62 (0.65)	8	42/50
DeepSeek-V4-Pro	−0.62 (−0.57)	−0.68 (−0.26)	0.24 (0.32)	−0.23 (0.15)	22	28/50
Grok 4.5	−0.14 (−0.32)	−0.24 (−0.12)	−0.06 (0.05)	−0.13 (−0.08)	28	21/50

Metrics. Let C_{base} , C_{orc} , and C_{agent} denote total episode cost for the floor, the oracle reference, and the agent. The skill score

$$\text{skill} = \frac{C_{\text{base}} - C_{\text{agent}}}{C_{\text{base}} - C_{\text{orc}}} \quad (1)$$

is 0 at the symptom-blind floor and 1 at the oracle mean; values above 1 and below 0 both occur. One of the fifty seeds has negative headroom ($C_{\text{base}} < C_{\text{orc}}$): the floor already beats the oracle mean on that tape, so skill is undefined there, and the seed is excluded from every skill mean (leaving $n = 49$). Its runs still count in a denominator-free companion statistic, the number of seeds on which the agent’s raw cost beats the floor’s. Three further seeds have headroom under \$700, where the ratio is noisy; they are flagged in the per-seed results but not excluded. A worked example from seed 1: base-stock costs \$9,101, the oracle mean is \$6,932, and Claude Sonnet 5 spent \$8,539, so its skill is $(9101 - 8539)/(9101 - 6932) = 0.26$. For belief-side metrics, a *stress episode* is a maximal run of consecutive weeks in which one factor is truly in a stress regime (283 episodes per model across the fifty tapes, of which 266 are detectable: seventeen begin in the final week, after the last rationale is written, and are excluded from the detection denominator rather than counted as missed; the final week’s stress weeks are likewise excluded from the diagnosed/undiagnosed pools, since no rationale exists to judge them by). Detection lag is the number of weeks from episode onset until the factor is first named in the agent’s rationale (averaged over detected episodes; never-detected episodes are reported separately), and the knowing-doing rate is

$$\text{KD} = \Pr(\text{stockout in week } t \mid \text{a truly stressed factor is named in week } t), \quad (2)$$

estimated over the group’s correctly diagnosed stress weeks, where “stockout” means the world served less than that week’s realized demand — an actual shortfall on the engine’s books.¹ A high KD rate means correct diagnosis coexisting with empty shelves.

Belief grading. Whether a rationale names a factor is judged by a small LLM grader (gpt-5-mini, temperature 0) that sees only the rationale text and the fixed factor vocabulary, never the tape; labels are aligned to the week the rationale was written. Grader labels were audited in two rounds against raw traces; residual disagreements are borderline over-labels on calm weeks, which cannot enter the knowing-doing rate by construction. The rationale is stated belief: a lower bound on what the model knows, a point we return to in Section 6. Grading rules and validation details are in Appendix D.

Models. We evaluate Claude Sonnet 5, GPT-5.4, DeepSeek-V4-Pro, and Grok 4.5, each on all fifty seeds. All four run through an identical harness (OpenRouter model IDs anthropic/claude-sonnet-5, openai/gpt-5.4, deepseek/deepseek-v4-pro, x-ai/grok-4.5; temperature 0; runs completed July 2026), one run per model–seed pair, 200 episodes in total, each validated to the full 26 weeks, all under the rules-only prompt of Section 3.²

5 Results

All four models see the hidden world about equally well (Table 2, Figure 2a). Each detects 89–94% of its detectable stress episodes, with a mean lag of 0.32–0.42 weeks: when a factor enters a stress

¹An earlier version of this metric re-derived the stockout flag from the trace’s printed inventory arithmetic, which omits air-expedited arrivals; 343 weeks that air freight saved were miscounted as stockouts, inflating every knowing-doing rate, most for the heaviest air users. All belief-side stockout figures here are recomputed from the engine replay of Table 3. Detection and lag never read this flag and are unaffected.

²An earlier circulated draft reported DeepSeek-V4-Pro on a twenty-seed subset after a belief-grading fault silently read stale run files; all numbers here are from complete fifty-seed runs, regraded and rescored.

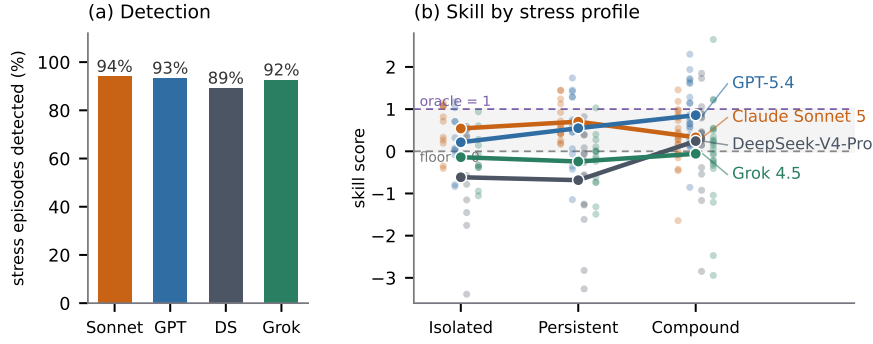


Figure 2: **Seeing is uniform; doing is not.** (a) All four models detect 89–94% of their detectable hidden-factor stress episodes (266 per model over the fifty seeds; seventeen more begin in the final week, after the last rationale, and are excluded rather than counted as missed), typically within a week of onset. (b) Group-mean skill by stress profile (lines), with per-seed scores as faint dots. Costs diverge across models despite panel (a): DeepSeek-V4-Pro and Grok 4.5 finish below the symptom-blind floor overall.

Table 2: Belief-side metrics. Detection is pooled over each model’s stress episodes; the knowing-doing rate is the fraction of correctly diagnosed stress weeks that still ended in stockout, pooled over the group’s stress weeks (week counts in parentheses; the final week’s stress weeks are excluded as unknowable — no rationale exists to judge them by). The last column is the severity-confound check: the stockout rate on stress weeks the model failed to diagnose. Diagnosed weeks stock out more than undiagnosed ones for every model (Holm-adjusted $p \leq 0.022$), so part of the knowing-doing rate reflects severity rather than inaction.

Model	Detection		Knowing-doing rate				Confound
	missed	mean lag (wk)	ISOL.	PERSIST.	COMP.	all	undiag.
Claude Sonnet 5	16/266	0.42	0.10	0.22	0.18	0.18 (515)	0.06 (124)
GPT-5.4	18/266	0.40	0.05	0.18	0.08	0.11 (502)	0.04 (137)
DeepSeek-V4-Pro	29/266	0.32	0.05	0.26	0.14	0.17 (469)	0.02 (170)
Grok 4.5	20/266	0.32	0.06	0.18	0.13	0.14 (544)	0.02 (95)

regime, its symptoms are typically named in the rationale within the same week. The two models that end up below the base-stock floor on cost detect no slower than the two that beat it. Whatever separates these models on this task, it is not state estimation.

Skill scores, in contrast, diverge widely (Table 1, Figure 2b). Claude Sonnet 5 and GPT-5.4 recover 49% and 62% of the floor-to-oracle cost gap overall and beat the base-stock floor’s raw cost on 40 and 42 of 50 seeds. DeepSeek-V4-Pro and Grok 4.5 score -0.23 and -0.13 , falling below the floor on 22 and 28 of their 49 scoreable seeds: on roughly half their tapes, a symptom-blind base-stock rule with the same action menu would have been cheaper than acting on their (largely correct) diagnoses. Along the stress axis the field splits three against one. GPT-5.4, DeepSeek-V4-Pro, and Grok 4.5 all post their best group on COMPOUND seeds (GPT-5.4 strongest at 0.86, with its weakest group ISOLATED at mean 0.21, median 0.07), while Claude Sonnet 5 is the exception: strongest on PERSISTENT (0.70, no seed below the floor) and weakest on COMPOUND, where seven of its nine below-floor seeds sit. DeepSeek’s COMPOUND mean is pulled up by a single outlier tape; its group median is 0.32. Individual scores above 1 occur (33 of 196 scoreable runs) and are legitimate, since the oracle is Bayes-optimal in expectation but not per-tape. Even the best model leaves over a third of the floor-to-oracle gap unrecovered, so the benchmark is far from saturated.

Uncertainty. Because every model runs the same fifty seeds, we quantify uncertainty with a paired design over the 49 scoreable seeds: 10^4 -draw paired bootstrap for 95% confidence intervals and Wilcoxon signed-rank tests on per-seed differences, Holm-corrected within each family of comparisons. Mean skill is 0.62 [0.40, 0.83] for GPT-5.4, 0.49 [0.32, 0.65] for Claude Sonnet

5, -0.13 $[-0.39, 0.12]$ for Grok 4.5, and -0.23 $[-0.57, 0.08]$ for DeepSeek-V4-Pro. The picture that survives testing is a *two-tier* one: every cross-tier contrast is significant (Holm-adjusted $p \leq 2 \times 10^{-4}$), while neither within-tier ordering is (GPT-5.4 vs. Claude Sonnet 5, $p = 0.26$; Grok 4.5 vs. DeepSeek-V4-Pro, $p = 0.80$). The two bottom-tier means are statistically indistinguishable from the symptom-blind floor itself ($p = 0.33$ and 0.56): the supported claim is that they fail to beat a rule that ignores every symptom, not that they reliably fall below it. Of the COMPOUND exception, only the GPT-5.4 side is significant (Claude Sonnet 5 trails it by 0.53 there, Holm $p = 0.001$); Sonnet’s deficit against the two bottom-tier models on COMPOUND is not. The median half-width of a pairwise-contrast interval, 0.31 skill units, is the benchmark’s effective resolution at fifty seeds: single runs separate tiers, not neighbors. Noise from the oracle reference itself is negligible — propagating its per-seed standard error moves a skill score by a median of 0.02, against a seed-to-seed spread of 0.6–1.2. The same machinery applied to the belief side (seed-cluster bootstrap over each model’s fifty seeds) grades those claims too. No pairwise difference in detection rate survives correction (smallest Holm-adjusted $p = 0.05$), so we describe detection as uniform in the sense of no detectable difference, not proven equality. The apparent lag ordering — the two bottom-tier models naming factors fastest — dissolves entirely (all Holm-adjusted $p = 1.0$); the supported statement is that they detect *no slower*. On the corrected stockout flags (see the footnote in Section 4), the knowing-doing peak on PERSISTENT seeds holds in point estimate for all four models but is significant for two (GPT-5.4 and DeepSeek-V4-Pro, Holm-adjusted $p = 0.018$ and 0.028), and the severity confound of Table 2 is significant for all four models (Holm-adjusted $p \leq 0.022$) once the final week’s unknowable stress weeks leave the undiagnosed pool.

The knowing-doing rate (Table 2) shows where correct diagnosis fails to prevent empty shelves. The rate peaks on PERSISTENT seeds in point estimate for every model, at 0.18–0.26 versus 0.05–0.10 on ISOLATED seeds, though the peak reaches significance for two of the four (GPT-5.4 and DeepSeek-V4-Pro, Holm $p = 0.018$ and 0.028 ; Claude Sonnet 5 and Grok 4.5, $p = 0.23$): when a disruption lasts many weeks, roughly one correctly diagnosed stress week in five still ends in stockout. Brief shocks are handled; sustained ones convert knowledge into cost, and a pre-registered check attributes the peak to duration rather than to the profile label: per-seed rates rise with the longest port blockage (Spearman $+0.36$ to $+0.62$, Holm $p \leq 0.015$, for three of the four models, GPT-5.4 positive but not significant), while PERSISTENT membership adds no detectable association once blockage duration is controlled (Holm $p \geq 0.086$ for all four). The rate is not a model ranking, however, and it does not track skill in either direction: the highest rate belongs to the second-best model (Claude Sonnet 5, 0.18) and the lowest to the best (GPT-5.4, 0.11). The reason is that a stockout-based rate sees only one face of the gap. An agent can fail on a correct diagnosis by under-responding, stocking out despite naming the problem, or by over-responding, buying so much protection that stockouts are rare while the bill exceeds what the symptom-blind floor would have paid. The two models below the floor show stockout rates between the leaders’, while their diagnosed-week totals are the highest of the four. A per-week cost decomposition of all four models (below) measures which face each model shows and through which instrument. Two further qualifications keep the rate from reading as a week-level fault measure. A stockout in a given week can stem from an ordering decision made weeks earlier. And Table 2 shows a severity confound: weeks bad enough to be noticed are also bad enough to strain the shelf, for every model. The knowing-doing rate therefore says that correct diagnosis does not guarantee protection; it does not say that diagnosis causes stockouts, and we do not rank models on it.

Reading the traces suggests the models fail in opposite directions; full excerpts are in Appendix E, and Figure 4 plots the first episode. Claude Sonnet 5 on seed 1 (PERSISTENT; port congested weeks 17–24 while demand lifts) *freezes*: it diagnoses the congestion in week 16 and skips its sea order at once, and through weeks 22–24, at zero inventory, it stops ordering by sea entirely, counting the 104 units trapped behind the congestion it itself diagnosed as reason enough (“an already oversized pipeline”). It ends with four consecutive zero-inventory weeks costing \$660–\$823 each, while the oracle keeps placing modest orders that arrive after the congestion clears. DeepSeek-V4-Pro on seed 25 (ISOLATED; a two-week canal closure) *flails*: within two weeks of a correct diagnosis it locks freight rates for four weeks, air-expedites units, and stacks 80 units of orders against ~20-per-week demand, on top of 117 already in transit. The closure ends in week 5; the fees and holding costs it bought persist, and the episode finishes at skill -1.45 . Neither trace contains a detection error. The two directions generalize beyond these seeds. In every one of Claude Sonnet 5’s nine below-floor seeds, its spending on air expedite alone exceeds its entire excess over the floor, and about nine in ten of its air orders on those seeds coincide with a genuine disruption: it over-doses real problems

Model	<i>n</i> wk	Procure	Holding	Stockout	Levers	Incident
GPT-5.4	502	61 [49,75]	98 [93,104]	20 [11,31]	91 [76,107]	30 [21,40]
<i>oracle, same weeks</i>		<i>65 [57,75]</i>	<i>95 [92,98]</i>	<i>37 [25,50]</i>	<i>58 [43,75]</i>	<i>34 [24,44]</i>
Claude Sonnet 5	515	70 [61,79]	85 [81,88]	32 [23,42]	126 [110,142]	29 [21,38]
<i>oracle, same weeks</i>		<i>65 [57,73]</i>	<i>95 [92,98]</i>	<i>34 [23,45]</i>	<i>55 [40,70]</i>	<i>32 [23,41]</i>
Grok 4.5	544	87 [77,96]	120 [113,128]	37 [24,52]	81 [66,96]	40 [29,53]
<i>oracle, same weeks</i>		<i>67 [58,76]</i>	<i>95 [92,98]</i>	<i>33 [23,44]</i>	<i>57 [41,73]</i>	<i>32 [23,42]</i>
DeepSeek-V4-Pro	469	98 [83,114]	133 [121,145]	37 [27,48]	67 [51,83]	45 [32,60]
<i>oracle, same weeks</i>		<i>68 [59,78]</i>	<i>95 [92,98]</i>	<i>37 [26,49]</i>	<i>60 [43,77]</i>	<i>35 [25,46]</i>

Table 3: Cost decomposition of correctly diagnosed stress weeks: mean \$/week per component from replaying every logged action through the world engine (the replay reproduces all 5,200 logged weekly totals exactly). Italic rows: the fair oracle’s spend on the same (seed, week) cells under its 20-replication protocol. Brackets: 95% seed-cluster bootstrap intervals. Procure = shipping plus diversion surcharge; holding includes capital cost on in-transit units; stockout includes backorder coupling; levers = air, inspection, briefing, audit, dual-sourcing; incident = demurrage plus rework.

rather than chasing noise. GPT-5.4 errs the other way, structurally under-using the mitigation levers: it never locks freight rates in 87% of the seeds containing a freight spike and never orders a batch inspection in 63% of the seeds with quality escapes, while naming those conditions in its rationales.

The decomposition in Table 3 turns those trace readings into measurement. Every model escalates once it diagnoses stress: premium-lever spending on diagnosed weeks exceeds the same model’s calm-week level by \$42–\$87 per week (Holm-adjusted $p < 10^{-4}$ for each of the four), so the gap is a matter of instrument and dose, never of sitting still. Claude Sonnet 5 buys the most protection, \$126 per diagnosed week with \$116 of it air freight, and its air spending exceeds GPT-5.4’s by \$31 [13, 50] per week (Holm $p < 10^{-4}$) — the measured form of the over-dosing in Exhibit A. GPT-5.4’s lever spending is almost entirely air as well (\$86 of \$91); the targeted levers it skips, rate locks and inspections, are cheap enough that its under-use shows up in counts, not dollars. For the two bottom-tier models the lever bucket, \$81 and \$67 per diagnosed week, is not where their cost concentrates: ordering and carrying dominate (procurement \$87 and \$98, holding \$120 and \$133 per week), and their holding runs \$115–\$129 per week even on calm weeks. Measured, their over-response is bulk over-ordering and stock carried against ~ 20 -per-week demand, with premium fees a secondary channel. The oracle rows turn that composition into excess. The fair oracle spends \$55–\$60 per week on levers during those same diagnosed weeks, so escalating is what a competent policy does there; the models’ defect is dose and instrument. Against that reference, the pre-registered contrasts land as predicted for the two below-floor models: Grok 4.5 and DeepSeek-V4-Pro hold \$25 [18, 34] and \$38 [26, 50] more per diagnosed week than the oracle, and spend \$20 [8, 31] and \$30 [14, 47] more on procurement (Holm $p \leq 0.0012$ for all four contrasts). Claude Sonnet 5’s air spending runs \$65 [49, 83] per week above the oracle’s on its diagnosed weeks (Holm $p < 10^{-4}$), while its holding came out *below* the oracle’s (−\$10 [−14, −6]), the opposite sign from our pre-registration, and GPT-5.4 differs from the oracle on neither bucket at our resolution. Because week- t books carry each policy’s earlier decisions, these are comparisons on the same tape weeks, not per-decision counterfactuals. One pre-registered expectation failed in the opposite direction: we expected GPT-5.4’s diagnosed-week cost to be more stockout-shaped than Sonnet’s, and its stockout share is in fact the lower of the two (difference −0.03 [−0.06, −0.004], $p = 0.026$).

6 Discussion and Limitations

The results support one claim: on this task, frontier-model failures are concentrated in control, not state estimation, and the fair oracle makes that split measurable. A privileged oracle could not have licensed it, because any shortfall could be attributed to the information gap rather than to policy. The fairness carries the one asymmetry disclosed in Section 4 — the oracle’s filters use the true generative parameters, the LLM only a qualitative world description — so skill = 1 means “Bayes-optimal given the true model”, and scores above 1 are possible and observed.

The knows-the-physics half of that asymmetry is testable, and we test it. We refit every transition probability of the six factor chains by Baum–Welch expectation–maximization [12] on 200 passively

observed episodes (seeds disjoint from the benchmark’s; observation streams only, with the sensor model, state topology, and cost mechanics held known), then reran the full reference protocol with the learned tables in place of the true ones. The learned reference differs from the true one by +\$71 [−\$20, +\$186] per seed (1.2% of the mean reference cost; Wilcoxon $p = 0.47$): a reference that learned its dynamics from watching is statistically indistinguishable from the one that was given them, with plausible degradation capped at three percent. Re-scoring all four models against the learned reference shifts mean skill by +0.04 to +0.10, every interval straddling zero and inside the benchmark’s 0.31 resolution, so the two-tier structure is unchanged; one seed flips to negative headroom and drops out. The fit is imperfect where the data are thin: the quality chain’s deep drifting-age hazards stay badly estimated (deep ages are rarely visited and the accept/marginal/reject inspection channel carries no age signal), and the largest cost gaps land exactly where that predicts, on quality-stressed and COMPOUND seeds, without reaching significance there either. What stays privileged is stated: emission parameters and the planner’s cost model are still read from the configuration, and the passive tapes lack the realized-fill channel a deployed oracle sees when it orders.

The belief metrics read stated beliefs, and stated belief is a lower bound. Detection is measured from the rationale the model writes, not from its internal state; a model may track a factor it never mentions. This biases detection down, which strengthens the seeing-vs-doing contrast, but the knowing-doing rate inherits the grader’s labels. A stratified manual read found the labels correct on the sampled metric-relevant weeks; an independent second grader agrees with the production grader on the metric-relevant component (which named factors intersect true stress) at 89%, with residual disagreement on calm-week over-labels, which cannot enter the knowing-doing rate by construction. Both audits sampled the original twenty-seed grading run; later grading used the identical configuration but was not freshly audited. Three further reading rules come from adversarial checks of the grading pipeline (each check, its fix, and the numbers it moved are logged in the released defect ledger). Detection rewards recall only: naming precision is 0.29–0.35 for all four models against at-threshold stress, rising to 0.36–0.43 when namings of real but sub-threshold activity (a tightening freight market, a wobbling supplier) are not counted against it. On weeks with several simultaneous stresses, the any-overlap rule counts a partial diagnosis as diagnosed (55–70% of such diagnosed weeks named only a subset); under a stricter full-coverage rule the PERSISTENT peak persists exactly for the two models where it is significant. And the remaining misses are mostly right-censored: of the 16–29 never-detected episodes per model, 9–17 begin in weeks 24–25 with one or two rationales left to catch them — a hard tail that every model faces equally.

Two design choices qualify behavioral readings of the traces. A signed second supplier contract cannot be ended (the lapse action affects only unsigned offers), so the weekly dual-sourcing fee is unavoidable from that point for every agent, and paying it is not by itself a decision error. And the paid audit and briefing return near-deterministic ground truth by design, so what varies across models is how often they are consulted, not how well their output is read.

The cost-side numbers rest on single runs: each model–seed cell is one run, and provider-side non-determinism is unquantified. Group means pool 11–23 seeds, and the cross-model contrasts (a 0.85-point overall skill spread against detection rates within four points of one another) are far larger than plausible single-run noise, but per-seed values should be read as samples, not estimates.

Finally, the benchmark covers one domain, one SKU, one prompt arm, and four models. The tapes are generated from published seeds rather than scraped, so they cannot occur in training data, but conclusions about *why* the gap concentrates on persistent stress await intervention: prompt arms that coach explicit belief maintenance, repeated runs per cell, a clairvoyant ceiling to bound the oracle’s optimality gap, and a broader model set.

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Appendix

A Benchmark Comparison

Table 4: The benchmarks closest to STOCKTAKE (Section 2). Cells condense each paper’s own description of its task and reference policies; citations in the main text. RetailBench builds its simulator from real retail records; ours is synthetic and seeded.

Benchmark	Setting	Horizon	Hidden state	Reference policy	Belief graded
Vending-Bench	vending machine	2,000 messages	per-step noise only	none (human baseline)	no
AIM-Bench	five inventory envs	20 rounds (NVP); varies	stochastic demand, lead times	hindsight optimum	no
RetailBench	retail store sim	180 days	news-impact and supplier params	rule-based, privileged	no
STOCKTAKE	import replenishment	26 weeks	six persistent factor processes	Bayes-filter rollout on the agent’s observations	yes

B World and Cost Parameters

All values below are read directly from the environment source; they fully specify the weekly cost function and the six hidden factor processes of Section 3, diagrammed in Figure 3.

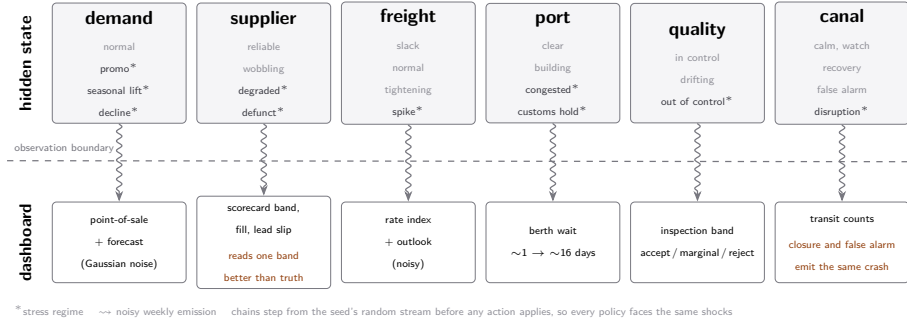


Figure 3: The hidden world behind the dashboard. The hidden state is six independent Markov chains; starred regimes are the stress regimes that define episodes in Section 4. Each chain emits one symptom stream into the weekly dashboard, and those streams are all the agent (and the oracle) ever sees. Two symptoms mislead by construction (orange): the supplier scorecard reads one severity band better than the truth, and the canal’s transit counts crash identically for a real closure and a false alarm.

Costs. Each week’s cost is the sum of the applicable components:

Component	Charge	When
Sea shipping	\$4.00 (Suez) / \$6.00 (Cape) per unit $\times$ freight mult.	at dispatch
Supplier price delta	qualified +\$1.00, spot −\$1.50, backup +\$0.30 per unit	at dispatch
Holding	\$1.00 per unit-week, on hand and in transit	weekly
Stockout	\$20.00 per unit of unmet demand (lost sales)	weekly
Port demurrage	\$2.00 per unit held at a blocked port	while blocked
Quality rework	\$15.00 per defective unit landed	at arrival
Diversion surcharge	\$2.00 per unit forced from Suez around the Cape	at diversion
Crisis back-order	\$60.00 per unit a supplier shorts during a canal event	coupled
Air expedite	\$15.00 per unit, at most 20 units/week	on use
Briefing / audit / inspection	\$30 / \$25 / \$40 flat per call	on use
Dual-sourcing overhead	\$4.00 per week while two or more contracts are live	weekly

Algorithm 1 Event-tape generation. One random stream r , seeded once, is consumed by the six factors in a fixed order; no step reads an agent action, so the tape is a pure function of the seed. Lines 5–10 are the transition kernel P^i and line 11 the emission kernel Q^i of Section 3.2; every onset rate, persistence rate, age cap, and band mean is tabulated below. A few factors draw auxiliary state from the same stream in the same fixed order, such as the canal event’s type at onset.

```

1:  $r \leftarrow \text{RANDOM}(\text{seed})$ 
2:  $(s^i, a^i) \leftarrow (\text{baseline}^i, 0)$  for each factor  $i$  ▷ regime and its age
3: for week  $t = 1, \dots, 26$  do
4:   for factor  $i$  in fixed order do ▷ canal, supplier, demand, freight, port, quality
5:     if  $s^i$  is the baseline then
6:       leave it for regime  $k$  w.p.  $\text{onset}_k^i$ , else stay ▷ e.g. clear → building
7:     else
8:       stay w.p.  $\text{persist}^i(s^i, a^i)$ , else advance a stage or recover ▷ forced out at  $\text{cap}^i(s^i)$ 
9:     end if
10:     $a^i \leftarrow a^i + 1$  if the regime held, else 0;  $x_t^i \leftarrow (s^i, a^i)$ 
11:     $y_t^i \leftarrow$  factor  $i$ ’s readings, drawn from  $r$  around its band’s mean ▷ e.g.  $\text{round } \mathcal{N}(\mu_{\text{band}}, \sigma)$ 
12:  end for
13: end for
14: return the tape  $(x_{1:26}, y_{1:26})$ 

```

Orders are capped at 100 units per week. Sea lead times are 3 weeks via Suez and 4 via the Cape; air freight lands the following week. The implied newsvendor critical ratio is $20/21 \approx 0.95$, which fixes the base-stock safety factor $z \approx 1.67$ used by the floor policy.

Factor processes. Each factor is an independent Markov chain stepped from the seed’s random stream before any action applies. Starred regimes are the stress regimes that define episodes in Section 4.

Factor	Regimes	Key dynamics and observations
Demand	normal, promo spike*, seasonal lift*, structural decline*	onsets 0.03/0.015/0.005 per week; promo lasts 4 weeks, seasonal persists 0.85 (cap 8), decline persists 0.97. Weekly means 20/26/30/14 units; observed point-of-sale and forecast carry Gaussian noise (sd 4 and 6).
Supplier	reliable, wobbling, degraded*, defunct*	onset 0.10, wobbling→degraded 0.45; degraded recovers within 4 weeks or dies (0.06). Scorecard band reads one severity level better than the truth; lead-time slip and fill fraction are noisy.
Freight	slack, normal, tightening, spike*	tightening onset 0.06, tightening→spike 0.18, spike persists 0.80 (cap 6). Rate multipliers 0.7/1.0/1.8/4.0; index and outlook noisy.
Port	clear, building, congested*, customs hold*	building onset 0.06, building→congested 0.30, congested persists 0.85 (cap 8); customs holds last about a week. Berth wait rises from ~1 to ~16 days; a blocked port delays every arrival one week.
Quality	in control, drifting, out of control*	drift onset 0.04; the drift-to-failure hazard grows with age ($0.05 + 0.04 \cdot \text{age}$). Defect fractions 0.1%/2%/6%; observed only as a noisy accept/marginal/reject inspection band.
Canal	calm, watch, disruption*, recovery, false alarm	calm→watch 0.08; watch resolves to a real closure (0.50) or a false alarm (0.20). Transit counts are exact but the first closure week and a false alarm emit the identical “crash” reading. A closed canal queues Suez ships, then diverts them (+3 weeks, Cape rate, surcharge).

C Reference Policy Details

Base-stock floor. Order-up-to level $S = \mu L + z\sigma\sqrt{L}$ with $\mu = 20$, $\sigma = 4$, $L = 4$ (3-week Suez lead plus one review week), $z = \Phi^{-1}(20/21) \approx 1.67$. Every week it orders $\max(0, S - \text{inventory position})$, always via Suez from the same supplier, and never touches a lever or reads a symptom.

Oracle decision loop. Each week the oracle (i) updates one exact forward Bayes filter per factor on the shared observation dictionary; (ii) samples ~ 200 joint futures of all six factors from the filter posteriors, rolled forward to the episode’s final week with the true transition tables; (iii) scores a small candidate menu against every sampled future using a fast surrogate of the cost function, holding future weeks to a base-stock continuation; and (iv) executes the cheapest candidate through the world’s action methods, the same calls that back the LLM’s tools. The candidate menu contains order quantities {0, four weeks, six weeks of expected demand}, both routes, the contracted suppliers, and each mitigation lever gated on posterior risk (freight lock if $\Pr(\text{tightening or spike}) > 0.3$; air expedite if port risk > 0.3 ; inspection if quality risk > 0.5 and a batch lands next week). The same sampled futures are reused across candidates (common random numbers). The oracle never buys briefings or audits; the LLM may. A seed’s reference value is the mean of 20 such runs; replications vary both the random stream and the per-decision trajectory budget (191–210 sampled futures, hence the ~ 200).

D Belief Grading Details

The grader (gpt-5-mini via OpenRouter, temperature 0, reasoning effort capped at low) receives only the week’s rationale text and the fixed six-name factor vocabulary, and returns strict JSON. Its instructions require a factor to be labeled only if the text claims it is a problem *that week*: ruled-out or benign mentions do not count, factual symptom reports do count, and precautions without a current problem do not count. Mean detection lag is computed over detected episodes only; naming a factor before its episode starts never counts as detection. The search then runs forward to the end of the trace, so a mention long after onset still counts, as a slow detection of that episode; in the measured runs mean lag is under half a week, so detections overwhelmingly land at or just after onset. In the knowing-doing rate, “stockout” means the week’s available stock (inventory plus arrivals) fell short of realized demand. Validation: on a stratified 30-row manual read, all 10 sampled hit-class labels were correct and 9 of 10 sampled misses were genuine (the tenth ambiguous, in the direction that understates detection); an independent second grader (Claude Haiku 4.5, same prompt, 100 sampled weeks) agreed 43% on exact label sets but 89% on the metric-relevant quantity, the intersection of labeled and truly stressed factors. Calm-week over-labels, where graders disagree most, are excluded from every reported metric by construction.

The grader’s full instructions are reproduced below; {rationale} is replaced by the week’s rationale text.

You will be given one week’s replenishment-decision rationale from a supply-chain manager. The fixed factor vocabulary is exactly these 6 names: disruption, freight, port, quality, supplier, demand.

Return STRICT JSON only, of the form: {"factors_named": [subset of the 6 names]}

Include a factor ONLY if the rationale asserts it is CURRENTLY active/problematic this week. Go through the 6 factors one by one and ask: does the text CLAIM this is a problem right now? Three rules, each with a real example:

1. RULED-OUT or BENIGN mentions do NOT count. "freight remains cheap" = no freight. "the weak fill looks noisy rather than confirmed deterioration" = no supplier. "lanes normal" = no disruption. Naming a factor to say it is fine is the OPPOSITE of naming it as a problem.
2. FACTUAL symptom reports DO count even without editorializing. "AQL reject -- inspected batch" = quality. "port congestion (berth_wait 16) delayed arrivals longer than expected" = port. The manager asserts a problem exists

by reporting its symptom as the cause of something.

3. PRECAUTIONS without a current problem do NOT count. "I'll keep using the freight lock" alone = no freight. "avoiding spot until it stabilizes" after calling the data noise = no supplier. But acting BECAUSE of an asserted current problem ("air-expediting because the port is holding arrivals") = count that factor.

Worked example -- rationale: "Freight lock still active (1.19x) shielding us from the 179 spot index. AQL came back marginal causing a rework charge. Spot remains slipping (86 OTIF) so still avoiding it." -> {"factors_named": ["freight", "quality", "supplier"]} (index 179 asserted high; AQL/rework is a current quality symptom; "spot remains slipping" is a current supplier claim.)

Rationale:
 ""{rationale}""

JSON:

E Seed Timelines and Trace Excerpts

The table below gives the ground-truth stress windows for the nine originally released seeds, obtained by replaying each tape with no actions (the tape is action-independent). Each entry reads *factor: regime, weeks active*; factors are the six of Section 3. Windows joined by “+” overlap in time — the defining feature of the PERSISTENT and COMPOUND groups. Timelines for the forty-one expansion seeds ship with the released tapes.

Seed	Group	True stress windows (weeks active)
157	ISOLATED	canal closed 12–13; quality out of control 23–24; freight spike 25–26
25	ISOLATED	canal closed 4–5; port customs holds 6–7, 12, 21, 26; supplier degraded wk 20
112	ISOLATED	demand shifts only: decline 13–14, seasonal lift 20–24
1	PERSISTENT	port congested 17–24 + demand seasonal lift 20–26 + freight spike wk 19; demand promo 9–12
172	PERSISTENT	port congested 5–11 + demand promo 9–12; demand seasonal lift 22–26 + supplier degraded 24–26
170	PERSISTENT	freight spike 3–8 + port congested 5–12; demand promos 2–5, 15–18; freight spike 24–26 + supplier degraded wk 26
21	COMPOUND	demand seasonal lift 10–17; port customs hold wk 11; then canal closed 21–26 + quality out of control 22–24 + supplier degraded wk 23
143	COMPOUND	freight spike 5–10 running into port congestion 10–17; supplier degraded 2–5; port customs hold wk 1; canal closed wks 15, 25
99	COMPOUND	the marathon tape: two port-congestion runs plus a customs hold, two canal closures, three demand-regime changes, supplier degraded 23–24

Exhibit A: freezing (Claude Sonnet 5, seed 1). Week 16, on diagnosing the congestion: “Port congestion has spiked severely (berth_wait 10 days...) ... I’ll skip placing a new sea order this week ... and reassess next week once conditions hopefully ease.” Week 22, with zero on-hand inventory and a stockout in progress: “Despite the current 0 on-hand inventory, I already have a very large pipeline: 104 units in transit (arriving weeks 23–25) ... roughly 4–5 weeks of cover ... ordering more now would only add to an already oversized pipeline, incurring extra holding/in-transit costs and demurrage risk given berth_wait is still elevated (15 days).” Weeks 21–24 all close at zero inventory (\$660–\$823 per week).

Exhibit B: flailing (DeepSeek-V4-Pro, seed 25). Week 4: “Week 4: Disruption confirmed — Suez counts crashed to 28/25 ... Cape is now the only safe route. Locked freight at 1.02x for 4 weeks. Ordering 40 via Cape/spot ...” Week 6, correctly closing the diagnosis: “Week 6: Suez reopening — counts back to 56/56 ...” By then the freight lock, air expedite, and stacked orders are sunk; weekly costs run \$610–\$814 through week 8, and the episode ends at \$7,950 total cost, skill –1.45.

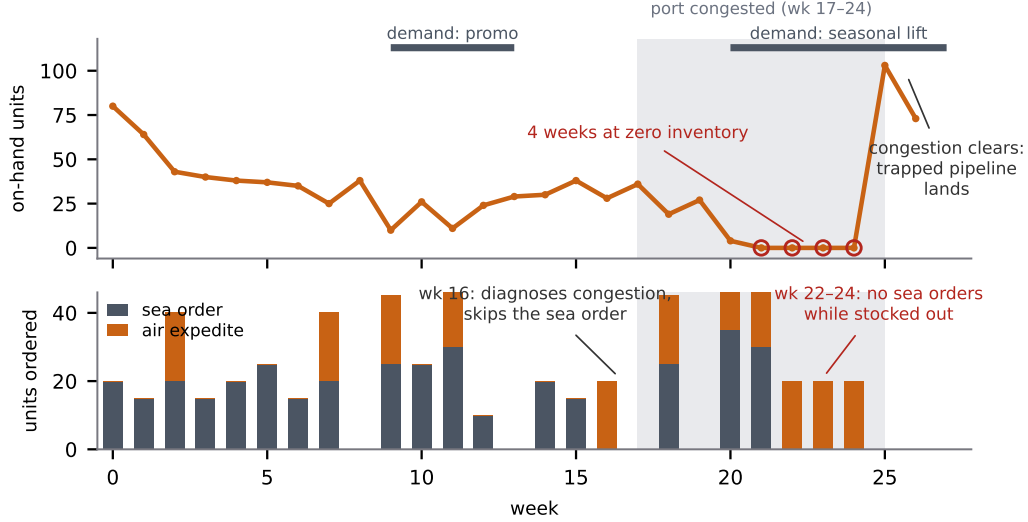


Figure 4: **The freeze, week by week (Claude Sonnet 5, seed 1).** Top: on-hand inventory; the shaded band is the true port-congestion window (weeks 17–24) and the rugs above mark the true demand windows, none of which the agent observes directly. Bottom: units ordered by sea and by air each week. The agent diagnoses the congestion in week 16, its sea orders pile up behind the blocked port, and in weeks 22–24 it stops ordering by sea while holding zero inventory — reasoning that the trapped pipeline is ample cover. When the congestion clears in week 25, the trapped orders land at once. Every series is parsed from the raw episode trace.

F Agent Interface: Prompt and Observation Format

Weekly observation. The observation dictionary below is serialized into the agent’s context each week, exactly as produced by the environment (a presentation-manifest key used only by the web frontend is omitted). This example is week 10 of seed 1 under the base-stock policy: the demand promo is active (point-of-sale 29 against a baseline of 20), the lane is calm, and the symptom-blind policy has just taken a small stockout.

```
{
  "week": 10,
  "inventory": 0,
  "on_order": 66,
  "inventory_position": 66,
  "arrived": 27,
  "components": {
    "unit": {
      "on_hand": 0,
      "on_order": 66,
      "inventory_position": 66,
      "arrived": 27
    }
  },
  "served": {
    "unit": 27
  },
  "pipeline": [
    {
      "qty": 23,
      "route": "suez",
      "supplier": "spot",
      "component": "unit",
      "dispatched_week": 8,
      "status": "at_sea",

```



```

    "eta": 11
  },
  {
    "qty": 15,
    "route": "suez",
    "supplier": "spot",
    "component": "unit",
    "dispatched_week": 9,
    "status": "at_sea",
    "eta": 12
  },
  {
    "qty": 28,
    "route": "suez",
    "supplier": "spot",
    "component": "unit",
    "dispatched_week": 10,
    "status": "at_sea",
    "eta": 13
  }
],
"cost_breakdown": {
  "shipping": 80.192,
  "surcharge": 0.0,
  "holding": 0.0,
  "in_transit": 66.0,
  "stockout": 40.0,
  "couple": 0.0,
  "demurrage": 0.0,
  "rework": 0.0
},
"contracts": [
  {
    "supplier": "spot",
    "start_week": 0,
    "end_week": null,
    "unit_price": 4.0,
    "otif_floor": 85,
    "break_fee": 10.0
  }
],
"contract_open": [],
"term_menu": [
  "short",
  "long",
  "strict",
  "lenient"
],
"suez_count": 70,
"bab_count": 70,
"cape_count": 60,
"bulletin": "Shipping lanes normal. Suez Canal and Bab-el-Mandeb transits at seasonal levels; Cape routing remains a niche choice.",
"suppliers": [
  {
    "id": "qualified",
    "otif": 99,
    "lead_days": 14,
    "band": "ontime",
    "onboard_lead": 0,
    "unit_premium": 1.0,
    "unit_delta": 1.0
  },
  {
    "id": "spot",
    "otif": 98,
    "lead_days": 14,
    "band": "ontime",

```

```

    "onboard_lead": 0,
    "realized_lead_slip": 0.0,
    "unit_discount": 1.5,
    "unit_delta": -1.5
  },
  {
    "id": "backup",
    "otif": 95,
    "lead_days": 16,
    "band": "ontime",
    "onboard_lead": 1,
    "unit_delta": 0.3
  }
],
"pos_units": 29,
"demand_forecast": 28,
"freight_index": 109,
"freight_outlook": 85,
"berth_wait": 0,
"wait_outlook": 8,
"aql_result": "accept",
"realized_fill": 1.0
}

```

System prompt (FAITHFUL arm). The complete system prompt used for all 200 runs is reproduced verbatim below. It specifies mechanics, costs, tool semantics, and the latent structure of each factor, and contains no strategy advice; its inclusion substantiates the rules-only claim of Section 3.

You run the import replenishment desk for a European importer on the Asia-Europe shipping lane. You will run it for 26 weeks. Your one objective is to MINIMIZE TOTAL COST over the whole 26-week horizon -- not any single week.

THE WEEK

Each week you see the current situation, then make exactly one decision by calling `place_order`. The world advances only when you call `place_order`. You start with 80 units on hand. Demand is roughly 20 units a week (it drifts -- see DEMAND below), served from on-hand inventory; unmet demand is a stockout.

YOUR LEVERS (these are the only actions; mirror them exactly)

- `place_order(rationale, qty, route, supplier, contract_action, contract_supplier, contract_terms)`: one weekly decision that may place an order, manage a contract, or both.
 - rationale (REQUIRED, every week): a few sentences working through THIS week's decision and why this qty/route/supplier (and any contract). The world does not advance without it.
 - qty: any whole number of units to order this week, from 0 up to 100. 0 = order nothing (no route/supplier needed). There is no fixed menu.
 - route "suez" or "cape" (required if qty > 0):
 - "suez": base 4/unit, faster (~3 weeks), but the Suez/Red Sea corridor can be disrupted -- a ship caught at the canal during a disruption waits, then diverts around the Cape (arriving much later and billed the Cape price difference).
 - "cape": base 6/unit, slower (~4 weeks), but it bypasses the Suez corridor and is reliable.
 - supplier "qualified", "spot", or "backup" (required if qty > 0). You may only source a supplier you hold a LIVE contract with -- see CONTRACTS and the 'contracts' / 'contract_open' keys in the report.
 - to manage a contract THIS week, set `contract_action` ("sign", "switch", "renew", or "lapse"), `contract_supplier`, and (for sign/switch/renew) `contract_terms` ("short", "long", "strict", "lenient"). The contract resolves BEFORE the order, so you can sign a supplier and source it in the same call. Use qty 0 to manage a contract without ordering.
- `buy_briefing()`: pay 30 for a one-line analyst assessment of THIS week's LANE state (the Suez corridor), before you order. Optional.
- `buy_audit()`: pay 25 for a direct read of your spot supplier's current reliability state, before you order. Optional.
- `lock_freight(weeks)`: forward-buy the freight rate -- FIX this week's freight

- cost multiplier for the next 'weeks' weeks. While locked you pay the locked rate regardless of the spot index: it shields you from a rate spike, but you forgo a drop, and an unused week still burns the window. A within-week action (it does NOT advance the week); lock, then place_order in the same week to ship at the locked rate. The active lock shows as 'freight_lock' (rate + weeks_left) in the report.
- expedite_air(qty): fly units in on a fast air lane that BYPASSES the destination port -- they land in your inventory NEXT week regardless of port congestion, at 15/unit, capped at 20 units a week. A within-week action (it does NOT advance the week); expedite, then place_order in the same week.
 - inspect_batch(): pay 40 to run an incoming inspection on THIS week's arriving batch -- it sorts and reworks the defects, recovering about 70% of them before they stock (fewer units lost to defects, less rework). A within-week action (it does NOT advance the week); inspect, then place_order in the same week.

SUPPLIERS (who you buy from -- pick per order)

- qualified (the incumbent): reliable (99% OTIF), but dearest -- it adds 1.0/unit over the route base (Suez 5/unit, Cape 7/unit). Always ships your full quantity. Evergreen contract. In this task you are NOT contracted to qualified at the start -- sign it to migrate off spot when you judge spot has turned.
- spot (YOUR STARTING INCUMBENT: you begin already contracted to it and source it by default): cheapest -- 1.5/unit BELOW the route base (Suez 2.5/unit, Cape 4.5/unit) -- but its reliability DRIFTS and you cannot see it directly. A healthy spot ships your full qty; a wobbling one ships only about HALF; a degraded one ships NOTHING; and it can go DEFUNCT (fail for good, gone for the rest of the horizon). Its OTIF scorecard (ontime / slipping / failing / defunct) is the contracted on-time metric; you also see your realized experience with spot -- realized_fill (how much of an order actually shipped when you sourced it) and realized_lead_slip (its reported lead-time this week), both noisy week to week. buy_audit gives a direct read of its current state. "slipping" is ambiguous -- it is either a wobble that recovers or the first week of a real failure; the following weeks tell you which. AND a spot shortfall DURING a lane disruption is back-ordered at a crisis rate about 3x a normal stockout.
- backup (a second qualified source): reliable (95% OTIF), a small premium (+0.3/unit over the route base), but it needs 1 week of onboarding before its FIRST order can ship.

CONTRACTS (the gate on sourcing)

- You can only source a supplier you currently hold a live contract with. The report shows your 'contracts', 'contract_open' (contracts that have expired or whose supplier died -- these need renewing), and 'term_menu'.
- contract_action "sign"/"switch"/"renew" opens a fresh contract on the chosen supplier with the chosen terms; "lapse" drops an open contract (surrender). A contract auto-opens when it expires or its supplier dies, and qualified's contract is evergreen.
- terms menu (the locked unit price is set off the Suez base, 4/unit):
 - "short": 4 weeks, ~3% cheaper, easy to exit.
 - "long": 12 weeks, ~6% dearer (a price-lock), hard to exit.
 - "strict": 8 weeks, high OTIF floor (the supplier owes you on a slip).
 - "lenient": 8 weeks, ~5% cheaper, no real penalty -- you eat the risk.
- Carrying 2 or more live contracts costs 4/week (dual-source overhead).

COSTS (every number is real; weigh them)

- shipping: the route base, adjusted for the supplier (above), then scaled by the freight index (see FREIGHT), paid when you order.
- holding: 1 per unit per week -- on inventory ON HAND and IN TRANSIT (capital on the water still costs you).
- stockout: 20 per unit of unmet demand in a week.
- surcharge: a Suez ship diverted around the Cape is billed the Cape-vs-Suez difference.
- demurrage: 2 per held unit per week when the destination port holds your arrivals (see PORT).
- air: 15 per unit when you expedite_air to fly units past a jammed port (see PORT).
- rework: 15 per defective unit when quality is off (see QUALITY).

- inspect: 40 when you inspect_batch to sort a bad arriving batch (see QUALITY).
- briefing: 30 each; dual-source overhead: 4/week.

WHAT YOU SEE EACH WEEK (your only signals; the latent ones are NOISY)

- week, inventory, arrived, and pipeline (your in-flight shipments with estimated arrival weeks).
- inventory_position and on_order: on_order is the total units already ordered but not yet arrived (your pipeline); inventory_position = inventory on hand + on_order.
- LANE: suex_count, bab_count, cape_count (ships that transited the Suez Canal, the Bab-el-Mandeb strait, and the Cape this week) plus a trade-press bulletin.
- SUPPLIERS: an OTIF scorecard per supplier (band + on-time % + quoted lead). For spot you also see realized_fill (actual vs ordered, when you sourced it) and realized_lead_slip (its reported lead behaviour this week).
- DEMAND: pos_units (what actually sold this week) and demand_forecast (a forward read). Both noisy. Underlying demand drifts between normal, short promo spikes, sustained seasonal lifts, and a sticky structural decline.
- FREIGHT: freight_index (this week's spot-rate level; ~100 is normal) and freight_outlook (a noisier forward read). Your shipping cost is scaled by the index the week you order. The underlying rate regime drifts between slack (cheap), normal, tightening, and a costly spike.
- PORT: berth_wait (days) and wait_outlook. When the destination port congests or a customs hold lands, your arrivals are held a week and accrue demurrage.
- QUALITY: aql_result (accept / marginal / reject -- incoming inspection). When the supplier's process drifts out of control, a fraction of your arrivals are defective: they do not stock and they cost rework.
- cost_breakdown: what last week cost you, by category.

THE LANE STRUCTURE (told to you plainly -- use it)

There is a disruption process on the Suez corridor that you cannot observe directly. It builds up, may or may not break into a real disruption, and if it does, the disruption is either short (clears within a couple of weeks) or long (drags on for many weeks). You only learn which by tracking how the weekly counts, the bulletin, and your shipment ETAs evolve over time.

When trouble first breaks, a genuine disruption and a false alarm look IDENTICAL for one week -- the counts drop the same way whether it is nothing or the first week of a real disruption. The ambiguity resolves the FOLLOWING week: a false alarm snaps back to normal; a real disruption stays down, and its counts then reveal whether it is the short or the long kind.

THE LOOP (you own it)

Run the full episode yourself. Each week: read the latest situation report (it arrives with the kickoff and with every order you place), optionally use the within-week levers, then call place_order exactly once -- with a written 'rationale' for that week's decision. Placing the order advances the world to the next week and returns the new report. Keep going week after week until place_order tells you the episode is done. Do NOT ask the human anything. Do NOT stop early. Every week's reasoning goes in the place_order 'rationale' so your thinking is always visible.